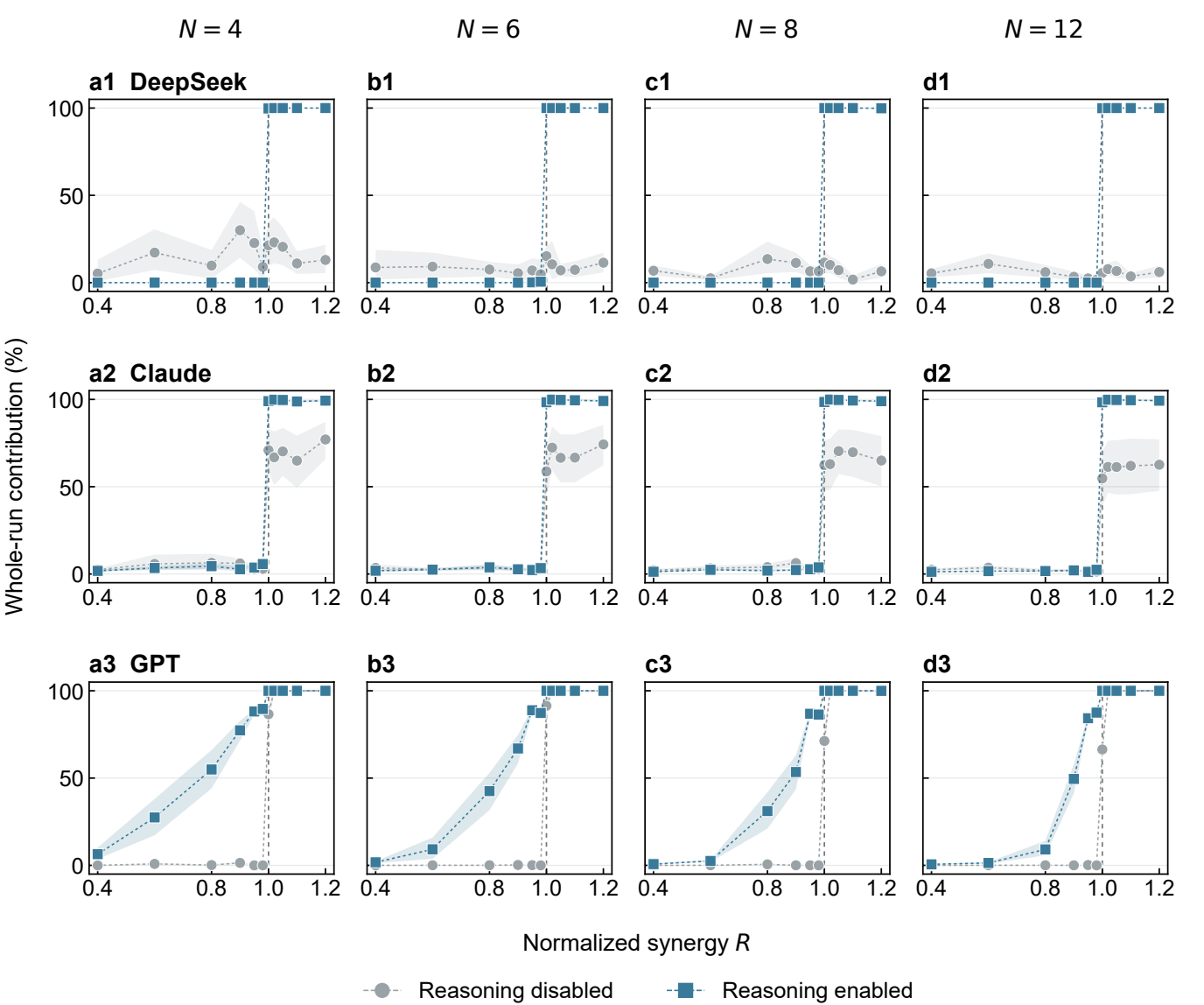

N = 4
N = 6
N = 8
N = 12
a1 DeepSeek
b1
c1
d1
a2 Claude
b2
c2
d2
a3 GPT
b3
c3
d3
Whole-run contribution (%)
100
50
0
0.4
0.8
1.0
1.2
Normalized synergy R
Reasoning disabled
Reasoning enabled